

Summary of Replication Package for “Evaluating State and Local Business Tax Incentives”

Cailin Slattery Owen Zidar
Columbia GSB Princeton University, NBER

May 7, 2020

This file contains all the data and code needed to replicate the exhibits and analysis in the main text of Slattery and Zidar’s article, “Evaluating State and Local Business Tax Incentives.” Please cite this paper if you use data and code; see `LICENSE.txt` for detailed permissions information.

Section 1 of this memo describes how to run code to replicate exhibits. Section 2 describes the data we use; subsection 2.1 describes pre-built data, and subsection 2.2 describes in brief the raw files we use in our main data build, as well as the way we process these raw files. Section 3 describes output, and section 4 gives formal citations for data sources discussed in section 2.

1 Running the code

To replicate the entire analysis, simply run the do-file entitled `make.do`, which will run all other do-files. In order to run this file, you will have to change the filepaths at the top of the `make.do` file, in the section labelled “SET FILEPATHS.” For instance, if you are running code on a Macintosh computer, you should change the global replication_root from “\$user/jep_firm_location/replication” to “/Downloads/replication,” assuming you have downloaded the replication file into your Downloads folder.

If you only want to run certain subsections of `make.do` or other do-files contained in the “do” folder, you will still have to set filepaths as described above, as well as install packages and run programs we’ve written by running the do-files stored in the “programs” subfolder. To accomplish all these tasks, it is sufficient to run the section of `make.do` entitled “PRELIMINARIES.”

The folder “do” contains three types of do-files:

1. Do-files which construct analysis data sets (stored in the “processed” subfolder of the “data” folder) from raw data sets (stored in the “raw” subfolder of the “data” folder). These are stored in the “build” subfolder
2. Do-files which construct exhibits, namely tables and figures, which are stored in the “tables” and “figures” subfolders, respectively
3. Do-files defining programs we have written to facilitate our databuild and analysis, stored in the “programs” subfolder

2 Data

2.1 Prebuilt data: Quarterly Workforce Indicators (QWI) averages

We use QWI averages of employment and payroll to supplement our data from the Quarterly Census of Employment and Wages (QCEW) data where QCEW data is missing. Specifically, we use NAICS 3-digit employment and wage data in the construction of the deal-specific analysis dataset if it is not available from the QCEW for a certain county-year-industry. This is the case for a small number of observations. The deal-specific analysis dataset is the underlying dataset for figures 2, 3, and 4 and table 4.

QWI data as downloaded by us via the QWI API on May 10, 2019 is stored in the “qwi_prebuild/raw” subfolder of the “data” folder. The subfolder “raw” of the folder of the folder “data/qwi_prebuild” has been zipped, because it contains over 100,000 (small) files. In order to read these files, unzip “raw.zip.” The resultant directory structure should be as follows: “qwi_prebuild/raw.” Our API call is also stored in the “qwi_prebuild” subfolder. The do-file `prebuild_qwi.do` in “do/build” builds the `qwi_avgs.dta` file in the “data/raw” folder. We describe `qwi_avgs.dta` in the following subsection. We “pre-build” the QWI data because it takes several hours to build the `qwi_avgs.dta` from our raw data.

2.2 Main data build

Data files are stored in the “data” folder. Raw data, stored in the “raw” subfolder of the “data” folder, is underlying data used in build do-files to construct processed data. The “raw” subfolder contains the following raw files:

- `acs_census_raw` [folder containing raw data]: Contains ACS 2010, Census 2000, and Census 2010 datafiles retrieved via Census API (<https://api.census.gov/data.html>) on April 15, 2019.
 - *ACS 2010 data*: ACS 1-year tables B06003 (Place of birth by sex in the United States), S1501 (Educational attainment), B25077 (Median home value - dollars), B25064 (Median gross rent - dollars), B25001 (Housing units). API call <https://api.census.gov/data/2010/acs/acs1>.
 - *Census 2000 data*: Decennial Census Summary Files 1 and 3. Tables DP1 (Profile of general population and housing characteristics) and H002 (Urban and rural) in summary file 1; tables P021 (Place of birth by citizenship status), QTP20 (Educational attainment by sex), H085 (Median value - dollars - for all owner-occupied housing units), H063 (Median gross rent - dollars), and H001 (Housing units) in summary file 3. API calls <https://api.census.gov/data/2000/sf1> and <https://api.census.gov/data/2000/sf3>, respectively.
 - *Census 2020 data*: Decennial Census Summary File 1. Tables DP1 (Profile of general population and housing characteristics) and H2 (Urban and rural). API call <https://api.census.gov/data/2010/dec/sf1>.
- `alt_xwalk_countyfips_190305.dta`: Sheet of county names, state fips codes, county-level fips identifiers, and state abbreviations. Cleaned version of US Census Bureau (1990-2017). Retrieved from <https://www.census.gov/programs-surveys/gov-finances/data/historical-data.html>.

- `bds_e_szsic_release.csv`: Sheet of establishment entry by size of establishment, sourced from Census Business Dynamic Statistics. Retrieved from http://www2.census.gov/ces/bds/estab/bds_e_szsic_release.csv.
- `CAINC1_ALL_STATES_1969-2017.csv`: Sheet of Annual Personal Income by County. Retrieved from <https://apps.bea.gov/regional/downloadzip.cfm> by selecting CAINC1 from drop-down menu under heading “Local Area Personal Income Accounts.”
- `comp_rev.dta`: Employee count, capital stock, revenue, gross profit, market value, state income taxes, and total income taxes from Compustat Industrial (2002-2016). Downloaded from Wharton Research Data Services (requires subscription). Retrieved from <https://wrds-web.wharton.upenn.edu/wrds/>.
- `county_unemp_1990-2017.dta`: Sheet of BLS employment statistics at county and year level, from Bureau of Labor Statistics (1990-2017a). Cleaned and appended. Retrieved from https://apps.bea.gov/iTable/index_regional.cfm.
- `CPI_1920_2018_20190305.xlsx`: CPI series for all urban consumers with based period 1982-1984 = 100. Bureau of Labor Statistics (1920-2018). Retrieved from <https://www.bls.gov/cpi/data.htm>.
- `DEC_10_SF1_GCTPH1.US05PR.csv`: Decennial Census Summary File 1 Table GCT-PH1 (Population, Housing Units, Area, and Density). API call <https://api.census.gov/data/2000/sf1>.
- `fips_statenames_xwalk.dta`: Crosswalk sheet between fips codes and state names as used in Zidar (2019). Accessible https://scholar.princeton.edu/sites/default/files/zidar/files/zidar_tcfw_replication.zip.
- `gdpstate_sic_all.csv`: Sheet of state-level GDP, GOS, compensation, subsidies, and taxes from Bureau of Economic Analysis (1963-1996). https://apps.bea.gov/iTable/index_regional.cfm
- `govexp_styear.dta`: Sheet of state-level spending variables, sourced from US Survey of State and Local Government Finances via Urban-Brookings Tax Policy Center (1977-2016a). Retrieved from <https://slfdqs.taxpolicycenter.org/pages.cfm>.
- `govrev_styear.dta`: Sheet of state-level revenue variables, sourced from US Survey of State and Local Government Finances via Urban-Brookings Tax Policy Center (1977-2016b). Retrieved from <https://slfdqs.taxpolicycenter.org/pages.cfm>.
- `HPI_AT_BDL_county.xlsx`: Sheet of county-year level HPI retrieved from Bogin et al. (2019); see https://www.fhfa.gov/DataTools/Downloads/Documents/HPI/HPI_AT_BDL_county.xlsx.
- `JVR_deals.dta`: Deals dataset from Bloom et al. (2019); retrieved from authors. See <https://www.aeaweb.org/articles?id=10.1257/aer.20170491>.
- `naics4d_labels.csv`: Four-digit North American Industry Classification System (NAICS) codes with descriptive names as used in Smith et al. (2019). See <https://scholar.princeton.edu/sites/default/files/zidar/files/syzz-capitalists.pdf>.

- `qcew_1990_2017_naics1d.dta`, `qcew_1d_long.dta`, `qcew_2d_long.dta`, `qcew_3d_long.dta`: Sheets of county- and year-level employment and wage information, retrieved from Bureau of Labor Statistics (1990-2017b). Retrieved from <https://www.bls.gov/cew/datatoc.htm>.
- `qwi_avgs.dta`: Sheet of average employment, firm count, and payroll at the county, 3-digit NAICS industry, and firm size level. Built as described in subsection 2.1.
- `SAGDP2N_ALL AREAS_1997_2017.csv`, `SAGDP3N_ALL AREAS_1997_2017.csv`, `SAGDP4N_ALL AREAS_1997_2017.csv`, `SAGDP5N_ALL AREAS_1997_2017.csv`, `SAGDP6N_ALL AREAS_1997_2017.csv`, `SAGDP7N_ALL AREAS_1997_2017.csv`: Sheets of state-level GDP, GOS, compensation, subsidies, and taxes from Bureau of Economic Analysis (1997-2017). Retrieved from https://apps.bea.gov/iTable/index_regional.cfm.
- `state_year_chars.dta`: Data set containing hand-collected state fiscal policy information, including total tax credits (revenue forgone via tax credits for businesses); total budget allocated to economic development programs for business; and the sum thereof. These variables are sourced from state tax expenditure reports and budget documents. In order to create these variables we download each tax expenditure report and budget document from state websites for the years 2007-2014. If those items are not available online, we contact the state Department of Revenue and/or Budget Office. We read each document to identify tax credits and budget items targeted at businesses.
- `taxrates_stateyear_1950_2017.dta`: Sheet of tax rates at state level. Sources of variables are as follows: corporate tax rates and sales tax rates from Book of States (see <http://knowledgecenter.csg.org/kc/category/content-type/content-type/book-states>); top personal income tax rate and capital gains income tax rates from NBER TAXSIM (Feenberg et al, 1993; see <https://users.nber.org/~taxsim/state-rates/>).

The do-files in the “build” subfolder of the “do” folder will output processed data, built from raw data files, to the “processed” subfolder of the “data” folder.

The firm-specific incentive deal dataset was collected for Slattery (2019). At this time, we will not provide all of the firm specific deal information (namely firm, names, subsidy size, jobs promised and investment promised). It will be publicly available with the publication of Slattery (2019), with a data appendix that details the sources for each deal. We have made collapsed versions of these datasets available in the “exhibit collapses” subfolder within “processed” subfolder under “data” to facilitate replication of all exhibits in the Slattery and Zidar (2020) paper. Replication files for Appendix Figure 1 are not available, as its reproduction requires the proprietary deal-level data.

3 Output

Running `make.do` will output exhibits into the “out” folder in this replication folder, which is organized into subfolders for tables and figures. Each of these subfolders have further subfolders entitled “appendix” which are meant to contain exhibits in the appendix of the article.

4 References:

- Bloom, Nicholas, Erik Brynjolfsson, Lucia Foster, Ron Jarmin, Megha Patnaik, Itay Saporta-Eksten, and John Van Reenen. 2019. “Data for: What Drives Differences in Management

Practices?” American Economic Review, 109 (5): 1648-83. See <https://www.aeaweb.org/articles?id=10.1257/aer.20170491>

- Bogin, Alexander N., Doerner, William M. and Larson, William D. (2019). “Data for: Local House Price Dynamics: New Indices and Stylized Facts”. *Real Estate Economics*, 47(2), 365-398. Accessed July 29, 2019. https://www.fhfa.gov/DataTools/Downloads/Documents/HPI/HPI_AT_BDL_county.xlsx
- Bureau of Economic Analysis. (1963-1996) “Gross Domestic Product, Gross Operating Surplus, Compensation, Subsidies, and Taxes.” Accessed February 20, 2018. https://apps.bea.gov/iTable/index_regional.cfm
- Bureau of Economic Analysis. (1969-2017). “Local Area Personal Income Accounts.” Accessed August 22, 2019. <https://apps.bea.gov/regional/downloadzip.cfm>
- Bureau of Economic Analysis. (1997-2017). “Gross Domestic Product, Gross Operating Surplus, Compensation, Subsidies, and Taxes.” Accessed February 20, 2018. https://apps.bea.gov/iTable/index_regional.cfm
- Bureau of Labor Statistics. (1920-2018). “Consumer Price Index - All Urban Consumers.” Accessed March 5, 2019. <https://www.bls.gov/cpi/data.htm>
- Bureau of Labor Statistics. (1990-2017a). “Local Area Unemployment Statistics.” <https://www.bls.gov/lau/#tables>
- Bureau of Labor Statistics. (1990-2017b). “Quarterly Census of Employment and Wages.” Accessed March 7, 2019. <https://www.bls.gov/cew/datatoc.htm>
- Compustat Industrial. (2002-2016). Available: Standard & Poors/Compustat. Accessed on July 31, 2019. <https://wrds-web.wharton.upenn.edu/wrds/>
- Feenberg, Daniel Richard, and Elizabeth Coutts (1993) “An Introduction to the TAXSIM Model.” *Journal of Policy Analysis and Management* 12(1), 189-194. <https://users.nber.org/~taxsim/state-rates/>
- Slattery, Cailin. (2019) “Data for: Bidding for Firms: Subsidy Competition in the U.S.” Working paper. <https://cailinlattery.com/research>
- Smith, Matthew, Owen Zidar, and Eric Zwick (2019) “Data for: Capitalists in the Twenty-First Century.” *Quarterly Journal of Economics* 134(4) 1675-1745. <https://scholar.princeton.edu/sites/default/files/zidar/files/syzz-capitalists.pdf>.
- Council of State Governments (1950-2017). “Book of the States.” <http://knowledgecenter.csg.org/kc/category/content-type/content-type/book-states>
- Urban-Brookings Tax Policy Center. (1977-2016a). “State and Local Government Finances: State Expenditures.” Accessed April 21, 2018. <https://slfdqs.taxpolicycenter.org/pages.cfm>
- Urban-Brookings Tax Policy Center. (1977-2016b). “State and Local Government Finances: State Revenues.” Accessed April 21, 2018. <https://slfdqs.taxpolicycenter.org/pages.cfm>

- US Census Bureau. (1977-2014). “Business Dynamic Statistics: Legacy BDS Establishment Characteristics Data.” Accessed July 30, 2018. http://www2.census.gov/ces/bds/estab/bds_e_szsic_release.csv
- US Census Bureau. (1990-2017). “Census of Governments: GOVS to FIPS Codes Crosswalk.” Accessed September 23, 2019. <https://www.census.gov/programs-surveys/gov-finances/data/historical-data.html>
- US Census Bureau. (1990-2018). “Quarterly Workforce Indicators.” Accessed May 10, 2019.
- US Census Bureau. (2000). “Decennial Census SF1; SF3.” Accessed April 15, 2019. <https://api.census.gov/data/2000/sf1>; <https://api.census.gov/data/2000/sf3>
- US Census Bureau. (2010). “American Community Survey, Place of birth by sex in the United States; Educational attainment; Median home value - dollars; Median gross rent - dollars; Housing units.” Accessed April 15, 2019. <https://api.census.gov/data/2010/acs/acs1>
- US Census Bureau. (2010). “Decennial Census SF1.” Accessed April 15, 2019. <https://api.census.gov/data/2010/dec/sf1>
- Zidar, Owen (2019) “Replication data for: Tax Cuts for Whom? Heterogeneous Effects of Tax Changes on Growth and Employment” *Journal of Political Economy* 127(3) 1437-1472. Accessible at https://scholar.princeton.edu/sites/default/files/zidar/files/zidar_tcfw_replication.zip